

INSIDER THREAT, IP THEFT & DATA EXFILTRATION INVESTIGATION PLAYBOOK

Document Control ID: DFIR-PB-002 **Target Standards:** ISO/IEC 27042:2015, NIST SP 800-86, CERT Insider Threat Guide **Case Identifier:** [INSERT_CASE_NUMBER] **Lead Examiner:** [INSERT_EXAMINER_NAME] | **Corporate Entity / Agency:** [INSERT_ORGANIZATION] **Subject Under Investigation:** [INSERT_EMPLOYEE_NAME_AND_USER_ID] **Investigation Classification:** HIGHLY CONFIDENTIAL // ATTORNEY-CLIENT PRIVILEGED

1. INTAKE, AUTHORIZATION & PRESERVATION PROTOCOLS

1.1 HR & General Counsel Coordination Checklist

- ☐ **Legal Authority Verified:** Written authorization received from Legal Counsel/HR prior to seizing or interrogating employee workstation/accounts.
 - ☐ **Preservation Hold Issued:** Legal hold notification served to IT Operations to suspend mailbox retention policies, cloud account auto-purges, and audit log roll-overs.
 - ☐ **Silent Account Suspension:** Coordinate timing so Active Directory, SSO (Okta/Entra ID), and VPN access are revoked simultaneously without alerting the subject during physical device acquisition.
 - ☐ **Mobile Device & BYOD Assessment:** Identify if company-managed MDM profiles (Intune/Jamf) or MAM containers reside on subject's personal mobile hardware.
-

2. REMOVABLE MEDIA & USB ARTIFACT HUNTING GUIDE

To establish physical exfiltration to flash drives, external SSDs, or SD cards, interrogate the following system registry hives and diagnostic logs in chronological sequence:

Investigative Question	Target Forensic Artifact / Key Path	Expected Value / Forensic Output
Was a USB drive connected?	HKLM\SYSTEM\CurrentControlSet\Enum\USB	Vendor, Product Model, Unique Serial Number, Device Class GUID.
First/Last time connected?	C:\Windows\INF\setupapi.devlog	Log local hardware installation date, first insert time, and driver initialization.
What drive letter did it get?	HKLM\SYSTEM\MountedDevices	Maps the unique Volume GUID (\\?\Volume{...}) to drive letter (e.g., E:\).
Which user plugged it in?	HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2	Specifies user profile and mapping proving user account active at mount.
Did user browse files on USB?	UsrClass.dat -> Local Settings\ShellBags\Microsoft\Windows\Shell\Bags	ShellBags are Microsoft Windows directory traversal and folder viewing on external volumes.

2.1 Removable Media Device Profile Summary Table

Device Identifier (Make/Model)	Device Serial #	Volume GUID	Assigned Drive Letter	First Insert Time (UTC)	Last Removal Time (UTC)	User Account Active
[INSERT_MAKE][INSERT_SERIAL_NUMBER][INSERT_VOLUME_GUID][INSERT_DRIVE_LETTER][INSERT_TIME_INSERT][INSERT_TIME_REMOVE][INSERT_USERNAME]						
\						

3. CLOUD STORAGE & WEB-BASED EXFILTRATION TRIAGE

3.1 Local Cloud Sync Client Database Inspection

Inspect local client sync SQLite databases to reconstruct batch file syncs prior to employee resignation:

- **Microsoft OneDrive:**

- SQLite DB Path: C:\Users\[USERNAME]\AppData\Local\Microsoft\OneDrive\settings\Personal\
- Target File: SyncEngineDatabase.db / ClientPolicy.ini
- Artifacts: Timestamped record of added files, modified timestamps, sync completion flags.

- **Google Drive for Desktop:**

- SQLite DB Path: C:\Users\[USERNAME]\AppData\Local\Google\DriveFS\[USER_ID]\
- Target File: metadata_sqlite_db
- Artifacts: Original item names, parent directory ID, cloud sync state, remote download URLs.
- **Dropbox:**
 - SQLite DB Path: C:\Users\[USERNAME]\AppData\Local\Dropbox\instance1\
 - Target File: filecache.db / sync_history.db

3.2 Web Browser Exfiltration & Staging Inspection

Search default browser profiles (Google Chrome, Microsoft Edge, Mozilla Firefox) located in C:\Users\[USERNAME]\AppData\Local\:

Browser Artifact	Database / File Location	Target Table / Field	Forensic Significance
Download History	... \Default\History	downloads table	Software downloads (e.g., VPNs, compression tools, file transfer agents).
URL Navigation	... \Default\History	urls / visits table	Access to personal webmail (Gmail/Proton), file drop sites (WeTransfer, Mega, Pastebin).
File Upload Headers	... \Default\Network Action Predictor	Upload records	Proves browser POST operations containing large payload byte counts to external URLs.

4. ARCHIVE CREATION, STAGING & INTELLECTUAL PROPERTY PARSING

4.1 Compression & Bulk Staging Tool Artifacts

Departing personnel frequently compress large directory structures (code repositories, CAD drawings, client lists) prior to exfiltration.

- **7-Zip History:** HKCU\Software\7-Zip\Compression and ExtractionPathHistory (Records target source folders and output archive paths).
- **WinRAR History:** HKCU\Software\WinRAR\ArcHistory and DialogEditHistory\ArcName (Records recently built .rar/.zip container names).
- **Target File Knowledge (LNK & Jump Lists):**
 - Path: C:\Users\[USERNAME]\AppData\Roaming\Microsoft\Windows\Recent\

- Parse .lnk and AutomaticDestinations-ms files for evidence of accessing staged archives or proprietary folders.

4.2 Staged & Compromised Intellectual Property Asset Log

File / Archive Name	Source Path on Host Disk	File (Bytes)	Size	SHA-256 Hash	Staged / Exfiltrated To (USB/Cloud/Web)	Timestamp (UTC)
SourceCode_Master_2025-01-15	C:\Users\USER\Documents	[INSERT_FILE_SIZE]	[INSERT_SHA256_HASH]	Drive E: (Sandisk Extreme USB)	[INSERT_TIMESTAMP]	
Client_Master_PrimaryShare	C:\ProgramData\MSERS	[INSERT_FILE_SIZE]	[INSERT_SHA256_HASH]	WebTransfer Web Upload	[INSERT_TIMESTAMP]	

5. ANTI-FORENSICS & EVIDENCE CONCEALMENT ARTIFACTS

Verify whether intentional evasion, file wiping, or log deletion occurred during the exfiltration window:

- ☐ **Audit Log Deletion:**
 - Windows Security Log Event ID **1102** (“The audit log was cleared”).
 - Windows System Log Event ID **104** (“The log file was cleared”).
- ☐ **Third-Party Sanitizer Execution:**
 - Check Prefetch (C:\Windows\Prefetch*.pf), Shimcache, and Amcache for execution traces of: CCLEANER.EXE, BLEACHBIT.EXE, BCWIPE.EXE, ERASER.EXE, SDELETE.EXE.
- ☐ **Volume & Journal Anomalies:**
 - Check NTFS \$UsnJrnl for massive clusters of USN_REASON_FILE_DELETE entries occurring within seconds.
 - Inspect \$MFT record numbers for anomalous gaps or timestomping indicators (\$STANDARD_INFORMATION timestamp earlier than \$FILE_NAME timestamp).

6. INVESTIGATIVE SUMMARY & CHAIN OF EVIDENCE ATTESTATION

6.1 Final Case Assessment & Exfiltration Summary

- **Total Confidential Files Accessed:** [INSERT_FILE_COUNT]
- **Total Volume of Data Exfiltrated:** [INSERT_DATA_VOLUME_MB_OR_GB]
- **Primary Vector Utilized:** [] USB Mass Storage [] Personal Cloud Sync [] Web Upload / Webmail [] Network Share
- **Evidence Tampering Detected:** [] YES [] NO

6.2 Examiner Sworn Attestation

I certify that the forensic examination documented in this playbook was executed objectively in accordance with ISO/IEC 27042 standards and peer-accepted digital forensic methodologies. The identified artifacts substantiate the exfiltration vectors and timelines set forth above.

Lead Forensic Examiner Signature: _____

Printed Name: [INSERT_EXAMINER_NAME], [CREDENTIALS]

Title: [INSERT_TITLE_OR_RANK]

Date Completed: [INSERT_DATE]